

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

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JCDG³ 2026

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Unfolding a 4-polytope

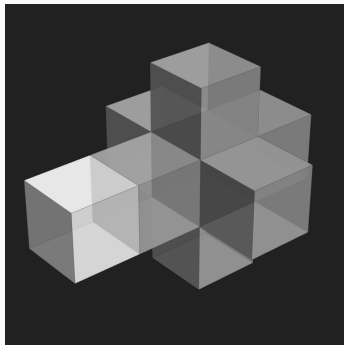
A **net** of a convex polytope: cut it open and lay the facets out flat, without overlap.

In 3D this is Dürer's problem, still open.

In 4D the facets are **3-dimensional cells** and the net lives in $\mathbb{R}^3$.

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): *every* spanning tree works for the 5-, 8-, 16-cell

Left open: 24-cell, 120-cell, 600-cell.



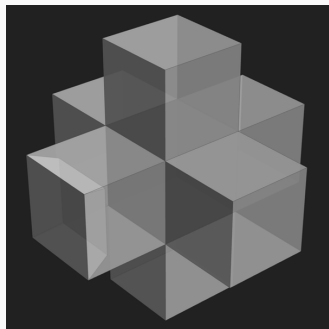
A net of the 8-cell: Dalí's cross

Face-rotation BFS

Fix a **root cell** r . Three steps.

- 1 **Face-up.** Rotate in $SO(4)$ so the centroid of r sits on the $+w$ axis.
- 2 **BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from r . For each tree edge (parent p , child c), rotate c about the shared polygon until it lands in p 's hyperplane.
- 3 **Project.** All cells now share one w -coordinate (spread $< 4 \times 10^{-13}$). Drop w .

The result is a net iff no two cells share interior volume.



Opening one BFS layer at a time, into Dalí's cross

Step 2 hides a choice

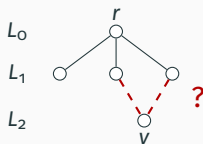
The layers are forced: L_d is the set of cells at distance d from r , and no rule can change that.

The **parent** is not. A cell in L_d may touch several cells of L_{d-1} , and BFS takes whichever one the implementation happens to reach first.

A geometric rule need not help

For the 5-, 8-, 16- and 24-cell the symmetries fixing r and v permute the candidates transitively, so **every** equivariant score — height along w , turning angle, the volume form — ties on all of them.

Ours breaks the tie by cell index, which is not equivariant either. **We come back to this.**



Two admissible parents
for v

The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(\mathbf{c}_i) \cap \text{conv}(\mathbf{c}_j)) \leq 10^{-8}.$$

Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

16-cell 0/16, 24-cell 0/24, 600-cell 0/600.

Measuring *volume* ignores contact of measure zero.

Main result

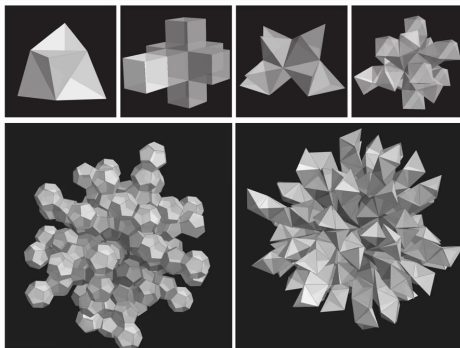
Theorem

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.

$5 + 8 + 16 + 24 + 120 + 600 = 773$
roots, all verified.

New beyond

Devadoss–Harvey: the 24-,
120- and 600-cell.



Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee).

Polytope	BFS valid	apple-peel (RZ)	
		orderings	nets
120-cell	all roots	1440/1440	0/1440
600-cell	all roots	0/2400	—

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

Consequence

Having a valid *ordering* and having a valid *net* are independent properties.

And more: the tie-breaking question

The abstract left this open:

“Whether validity extends to all BFS tie-breakings ... remains open.”

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell — and the trees can be counted and enumerated with no rule at all.

	trees / root	checked
5-cell	1	all
8-cell	6	all
16-cell	20,736	all
24-cell	32,768	all
120-cell	$\sim 1.8 \times 10^{44}$	20 random
600-cell	$\sim 1.8 \times 10^{65}$	20 random

1,132,661 trees, no self-intersection. For the first four the rule is irrelevant — including the 24-cell. And BFS is not a vacuous restriction: the 24-cell is **not** all-net (Seamons 2026) — some spanning tree does overlap.

△ Warning: this does not extend

Beyond the regular case the rule stops being harmless.

Grand antiprism, 320 cells

rule A valid from **all 320** roots

rule B valid from **no** root

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: another swings from 1.8% to 99.8% of its roots.

What regularity buys is not that some tree works, but that the choice does not matter.



Grand antiprism, one valid net

Open problems

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees — settled for the 24-cell (Seamons 2026), still open for the 120- and 600-cell.
- For the 120- and 600-cell even the BFS family is open: 10^{44} and 10^{65} trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets:

`github.com/takashi-randomwalker/apple-peel-4d`

Backup: “BFS tree” is bigger than what BFS returns

A traversal reaches a tree iff its parent choices admit a total order, i.e. iff the constraint digraph $p(v) \rightarrow q$ is acyclic.

	BFS trees	reachable	
8-cell	6	6	100%
16-cell	20,736	4,896	23.6%
24-cell	32,768	2,592	7.9%

So of the 786,432 trees verified for the 24-cell, only 62,208 are reachable by any traversal. The obstruction is a *cycle* in the constraints, not merely two cells sharing two candidate parents: the 16-cell has no such pair and still leaves three quarters of its trees unreachable.

Backup: which rules were compared

	queue	max-index	min-index
Runcinated 24-cell	0/240	0/240	0/240
Runcitruncated 600-cell	47/2640	2635/2640	2627/2640
Runcitruncated 120-cell	2570/2640	2640/2640	2151/2640
Grand antiprism	320/320	320/320	0/320

“queue” is the textbook BFS loop, neighbours scanned in index order.
“min/max-index” choose, for each cell, the smallest or largest indexed candidate parent.

Uniformly random trees are harsher still: of the 61 uniform polytopes valid from every root under the queue rule, 13 fail within 10 random trees, the grand antiprism among them — and every one of the four polytopes with 2,640 cells fails all ten, whether or not it is one of the 61.

Backup: the one that no rule could unfold

Runcinated 24-cell $t_{0,3}\{3, 4, 3\}$, 240 cells, 0/240 under all three index rules. Ask instead for a *symmetric tree*: a BFS tree invariant under $H \leq \text{Stab}(r)$ exists iff every cell has a candidate parent fixed by its stabilizer in H . Here $|\text{Aut}| = 2,304$, with two cell orbits, and the largest such H has order 3.

root orbit	$ \text{Stab}(r) $	$\max H $	H -invariant	random
48 (octahedra)	48	3	70/600	0/600
192 (triangular prisms)	12	3	83/600	0/600

The control is the point: unconstrained trees at those roots give nothing. Transitivity on each orbit then gives a valid net from **all 240 roots** — not a polytope without a net, but one whose nets no index rule finds.

(No conflict with slide 4: there the candidates form one orbit under the *full* stabilizer, so no invariant tree exists at all.)

Backup: what does *not* separate them

Cell degree fails both ways. The grand antiprism and the snub 24-cell have cells of degree 4 — the poverty that stops apple-peel on the 600-cell — yet are valid from every root, while the runcinated 24-cell has minimum degree 5 and is valid from none. Nor does high degree help: every prism has a cell of degree up to 92 and all 17 are fine, and the root-dependent runcitruncated 120-cell carries cells of degree 32.

Angular defect at an edge — 2π less the dihedral angles meeting there, zero exactly when those cells tile $\mathbb{R}^3$ around it. For the regular six it is one number, and it orders them *exactly* by how hard they proved to be: 148.4° , 90° , 77.9° , 31.6° , 10.3° , 7.4° . Among the uniform ones it breaks both ways: the omnitruncated 120-cell shares its minimum defect, 0.736° , with the root-dependent runcitruncated 120-cell and is valid from every root, while the runcinated 24-cell sits at 10.5° , above many that are. Weighting by depth or by size does not repair it.

Backup: the six polytopes

Polytope	Symbol	V	F	C	Cell	deg	depth	trees / root
5-cell	$\{3, 3, 3\}$	5	10	5	tetrahedron	4	1	1
8-cell	$\{4, 3, 3\}$	16	24	8	cube	6	2	6
16-cell	$\{3, 3, 4\}$	8	32	16	tetrahedron	4	4	20,736
24-cell	$\{3, 4, 3\}$	24	96	24	octahedron	8	3	32,768
120-cell	$\{5, 3, 3\}$	600	720	120	dodecahedron	12	5	1.8×10^{44}
600-cell	$\{3, 3, 5\}$	120	1200	600	tetrahedron	4	15	1.8×10^{65}

deg counts the cells sharing a 2-face; *depth* is the eccentricity of the cell-adjacency graph, the same from every cell. The layers are 1+4; 1+6+1; 1+4+6+4+1; 1+8+14+1; 1+12+32+42+32+1; and sixteen layers for the 600-cell.

The 24-cell row tells the two families apart: every BFS tree of it fits in 1+8+14+1, whereas the overlapping spanning tree of Seamon's puts 13 of its 24 cells deeper than their distance from the root.